IMO(38^{th}) – 1997

First Day

1. In the plane the points with integers coordinates are the vertices of unit squares. The squares are colored alternately black and white (as on chessboard). For any pair of positive integers m and n, consider a right angled triangle whose vertices have integer coordinates and whose legs, of lengths m and n, lie along edges of the squares. Let S_1 be the total area of the black part of the triangle and S_2 be the total area of the white part. Let $f(m,n) = |S_1 - S_2|$.
   (a) Calculate $f(m,n)$ for all positive integers' m and n which are either both even or both odd.
   (b) Prove that $f(m,n) \le \frac{1}{2}\max\{m,n\}$ for all m and n.
   (c) Show that there is no constants C such that $f(m,n) < C$ for all m and n.
2. The angle at A is the smallest angle of triangle ABC. The points B and C divide the circum circle of the triangle into two arcs. Let U be an interior point of the arc between B and C which does not contain A. The perpendicular bisector of AB and AC meet the line AU at V and W, respectively. The lines BV and CW meet at T. Show that AU = TB + TC.
3. Let $x_1, x_2, ..., x_n$ be real numbers satisfying the conditions $|x_1 + x_2 + ... + x_n| = 1$ and $|x_i| \le \frac{n+1}{2}$, $i = 1,2,...,n$. Show that there exists a permutation $y_1, y_2, ..., y_n$ of $x_1, x_2, ..., x_n$ such that $|y_1 + 2y_2 + ... + ny_n| \le \frac{n+1}{2}$.

Second Day

4. An $n \times n$ matrix whose entries comes from the set $S = \{1,2,3,...,2n-1\}$ is called as silver matrix if, for each $i = 1,2,...n,$ the ith row and the ith column together contain all elements of S. Show that:
   (a) There is no silver matrix for $n = 1997$.
   (b) Silver matrices exist for infinitely many values of n.
5. Find all pairs (a,b) of integers $a,b \ge 1$ that satisfies the equation: $a^{b^2} = b^a$.
6. For each positive integer n, let $f(n)$ denote the number of ways representing n as a sum of powers of 2 with non-negative integers exponents. Representations which differ only in the ordering of their summands are considered to be the same. For instance $f(4) = 4$, because the number 4 can be represented in the following four ways: 4; 2+2; 2+1+1; 1+1+1+1. Prove that for any integer $n \ge 3$,
$$2^{n^2/4} < f(2^n) < 2^{n^2/2}.$$

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